

Practical 2: State Management and Routing in React

Objective:-	To implement reactive state management using useState and multi-page navigation using React Router in the existing portfolio application.
Problem Statement:-	• Extend the portfolio application from Practical 1 by adding React Router. • Implement a navigation bar with at least 3 routes: Home, Projects, and Contact. • Each route must render a distinct component without a full page reload. • Add at least 2 useState variables used meaningfully one for toggling UI visibility and one for managing a form input on the Contact page. • The Contact page must include a controlled input that captures and displays user input in real time.
Implementation:-	Navbar:  Projects: <pre>import { useEffect, useMemo, useState } from 'react' import { Search } from 'lucide-react' import { FiGithub } from 'react-icons/fi' import { profile } from '../utils/data' import Section from '../ui/Section' import Reveal from '../ui/Reveal' import Spinner from '../ui/Spinner' import ErrorMessage from '../ui/ErrorMessage' import RepoList from '../ui/RepoList' const GITHUB_API_URL = `https://api.github.com/users/\${profile.githubUsername}/repos` const Projects = () => { const [repos, setRepos] = useState([]) const [loading, setLoading] = useState(true) const [error, setError] = useState(null) const [query, setQuery] = useState('') const [attempt, setAttempt] = useState(0) useEffect(() => { let ignore = false setLoading(true) setError(null) fetch(`\${GITHUB_API_URL}?per_page=30&sort=updated`) .then((res) => { if (!res.ok) throw new Error(`GitHub API responded with \${res.status}`) return res.json() }) }, [attempt]) }</pre>

```
    })
    .then((data) => {
      if (!ignore) setRepos(data)
    })
    .catch((err) => {
      if (!ignore) setError(err.message)
    })
    .finally(() => {
      if (!ignore) setLoading(false)
    })
  })

  return () => {
    ignore = true
  }
}, [attempt])

const sorted = useMemo(
  () =>
    [...repos].sort(
      (a, b) => (b.stargazers_count ?? 0) - (a.stargazers_count ?? 0)
    ),
  [repos]
)

const filtered = useMemo(
  () =>
    sorted.filter((repo) =>
      repo.name.toLowerCase().includes(query.trim().toLowerCase())
    ),
  [sorted, query]
)

return (
  <Section
    id="projects"
    eyebrow="Projects"
    title="Open-source work"
    description="A live list of my public repositories, fetched
straight from the GitHub API."
  >
    <Reveal>
      <div className="mb-6 flex flex-col items-start justify-between
gap-4 sm:flex-row sm:items-center">
        <p className="inline-flex flex-wrap items-center gap-1.5 text-
sm text-muted">
          <FiGithub className="size-4 text-bronze" />
          <span>
            {loading
              ? 'Fetching from GitHub...'
              : error
            }
          </span>
        </p>
      </div>
    </Reveal>
  </Section>
)
```

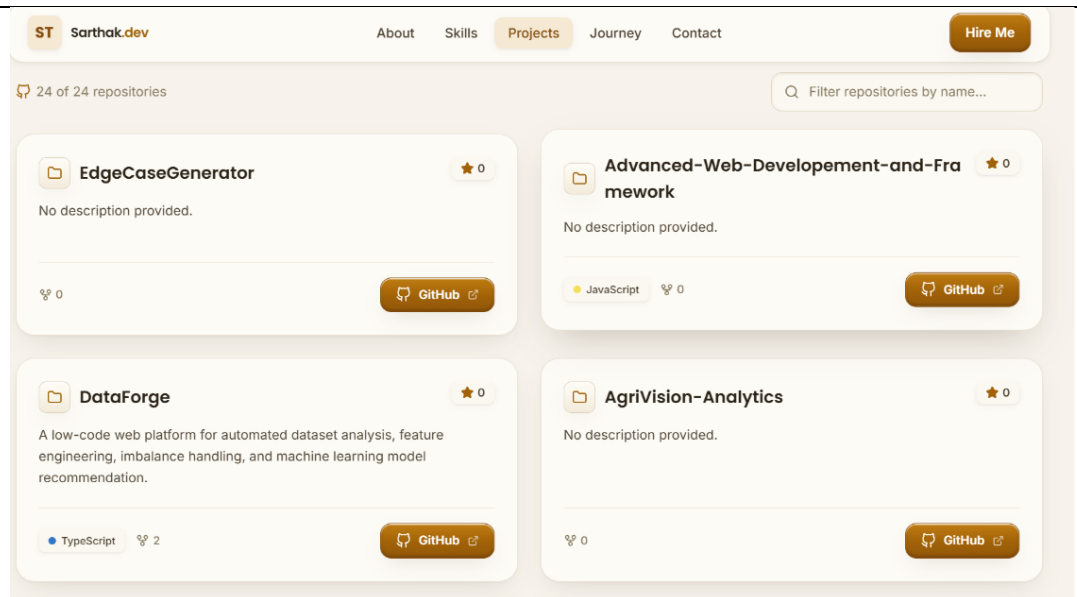
```

      ? 'Showing GitHub data'
      : `${filtered.length} of ${repos.length}
repositories${
      query ? ' matching your search' : ''
    }`
    </span>
  </p>
  <div className="relative w-full sm:w-72">
    <Search className="absolute top-1/2 left-3.5 size-4 -
translate-y-1/2 text-muted" />
    <input
      type="search"
      value={query}
      onChange={(e) => setQuery(e.target.value)}
      placeholder="Filter repositories by name..."
      aria-label="Filter repositories by name"
      className="w-full rounded-xl border border-line-strong bg-
surface py-2.5 pr-4 pl-10 text-sm text-ink shadow-
[inset_0_1px_3px_rgba(60,45,20,0.08)] outline-none transition-all
duration-200 placeholder:text-muted focus:border-bronze focus:bg-
parchment focus:ring-2 focus:ring-bronze/20"
    />
  </div>
</div>
</Reveal>

{loading ? (
  <Reveal>
    <Spinner />
  </Reveal>
) : error ? (
  <Reveal>
    <ErrorMessage
      message={error}
      onRetry={() => setAttempt((n) => n + 1)}
    />
  </Reveal>
) : (
  <Reveal>
    <RepoList repos={filtered} hasRepos={repos.length > 0} />
  </Reveal>
)}
</Section>
)
}

export default Projects

```



Contact:

```
import { useState } from 'react'
import { Send, Mail, Phone, CheckCircle2 } from 'lucide-react'
import { FiGithub, FiLinkedin } from 'react-icons/fi'
import { profile, socials } from '../utils/data'
import Section from '../ui/Section'
import Reveal from '../ui/Reveal'
import Button from '../ui/Button'

const contactCards = [
  { icon: Mail, label: 'Email', value: profile.email, href: `mailto:${profile.email}` },
  { icon: Phone, label: 'Phone', value: profile.phone, href: `tel:+919601039375` },
  { icon: FiGithub, label: 'GitHub', value: '@Talaviya-Sarthak', href: profile.github },
  { icon: FiLinkedin, label: 'LinkedIn', value: 'in/sarthak-talaviya', href: profile.linkedin },
]

const inputClass =
  'w-full rounded-xl border border-line-strong bg-surface px-4 py-3 text-sm text-ink shadow-[inset_0_1px_3px_rgba(60,45,20,0.08)] outline-none transition-all duration-200 placeholder:text-muted focus:border-bronze focus:bg-parchment focus:ring-2 focus:ring-bronze/20'

const Contact = () => {
  const [form, setForm] = useState({ name: '', email: '', message: '' })
  const [status, setStatus] = useState('idle')

  const update = (key) => (e) => setForm((f) => ({ ...f, [key]: e.target.value }))

  const onSubmit = (e) => {
```

```

    e.preventDefault()
    const mailto = `mailto:${profile.email}?subject=Portfolio enquiry
from ${encodeURIComponent(
    form.name || 'a visitor'
)}&body=${encodeURIComponent(form.message)}`
    window.location.href = mailto
    setStatus('sent')
    setForm({ name: '', email: '', message: '' })
    window.setTimeout(() => setStatus('idle'), 5000)
  }

  return (
    <Section
      id="contact"
      eyebrow="Contact"
      title="Let's build something"
      description="Have a role, a project or an idea? My inbox is always
open – I usually reply within a day."
    >
      <div className="grid gap-8 lg:grid-cols-[0.9fr_1.1fr] lg:gap-10">
        <div className="flex flex-col gap-5">
          {contactCards.map((card, idx) => (
            <Reveal key={card.label} delay={idx * 0.07}>
              <a
                href={card.href}
                target={card.href.startsWith('mailto') ? undefined :
'_blank'}
                rel="noreferrer"
                className="surface-raised group flex items-center gap-4
p-5 transition-all duration-300 hover:-translate-y-1 hover:shadow-hover"
              >
                <span className="btn-secondary inline-flex shrink-0
rounded-xl p-3">
                  <card.icon className="size-5 text-bronze-deep" />
                </span>
                <span>
                  <span className="block text-xs font-semibold tracking-
wide text-muted uppercase">
                    {card.label}
                  </span>
                  <span className="block text-sm font-medium text-ink
transition-colors group-hover:text-bronze-deep">
                    {card.value}
                  </span>
                </span>
              </a>
            </Reveal>
          )))
        <Reveal delay={0.25}>

```

```

        <div className="surface-raised p-5">
          <p className="text-sm font-semibold text-ink">Prefer quick
chat?</p>
          <p className="mt-1 text-sm text-muted">
            Find me on the platforms below, or drop a message
through the form.
          </p>
          <div className="mt-4 flex flex-wrap gap-2">
            {socials.map((s) => (
              <a
                key={s.label}
                href={s.href}
                target={s.href.startsWith('mailto') ? undefined :
'_blank'}
                rel="noreferrer"
                className="btn-secondary inline-flex items-center
gap-2 rounded-xl px-3.5 py-2 text-xs font-semibold transition-all
duration-200 hover:-translate-y-0.5"
              >
                {s.label}
              </a>
            ))}
          </div>
        </div>
      </Reveal>
    </div>

    <Reveal delay={0.1}>
      <form onSubmit={onSubmit} className="surface-raised p-6 sm:p-
8">
        <h3 className="font-display text-lg font-semibold text-
ink">Send me a message</h3>
        <p className="mt-1 text-sm text-muted">
          This opens your email client with the message pre-filled.
        </p>

        <div className="mt-6 grid gap-4 sm:grid-cols-2">
          <div>
            <label htmlFor="name" className="mb-1.5 block text-xs
font-semibold text-ink-soft">
              Name
            </label>
            <input
              id="name"
              type="text"
              required
              value={form.name}
              onChange={update('name')}
              placeholder="Jane Doe"
              className={inputClass}
            />
          </div>
        </div>
      </form>
    </Reveal>
  </div>

```

```

        />
      </div>
      <div>
        <label htmlFor="email" className="mb-1.5 block text-xs
font-semibold text-ink-soft">
          Email
        </label>
        <input
          id="email"
          type="email"
          required
          value={form.email}
          onChange={update('email')}
          placeholder="jane@company.com"
          className={inputClass}
        />
      </div>
    </div>

    <div className="mt-4">
      <label htmlFor="message" className="mb-1.5 block text-xs
font-semibold text-ink-soft">
        Message
      </label>
      <textarea
        id="message"
        required
        rows={5}
        value={form.message}
        onChange={update('message')}
        placeholder="Tell me about your project or opportunity..."
        className={` ${inputClass} resize-none`}
      />
    </div>

    <div className="mt-6 flex items-center gap-4">
      <Button as="button" type="submit" variant="primary">
        <Send className="size-4" />
        Send Message
      </Button>
      {status === 'sent' && (
        <p className="inline-flex items-center gap-1.5 text-sm
font-medium text-tick">
          <CheckCircle2 className="size-4" />
          Opening your mail client...
        </p>
      )}
    </div>
  </form>
</Reveal>

```

```
    </div>
  </Section>
)
}

export default Contact
```

CONTACT

Let's build something

Have a role, a project or an idea? My inbox is always open — I usually reply within a day.

 **EMAIL**
stalaviya709@gmail.com

 **PHONE**
+91 96010 39375

 **GITHUB**
@Talaviya-Sarthak

 **LINKEDIN**
in/sarthak-talaviya

Send me a message

This opens your email client with the message pre-filled.

Name

Email

Message

Tell me about your project or opportunity...



404 Error

```
import { Link } from 'react-router-dom'
import { motion, useReducedMotion } from 'framer-motion'
import { Compass, Home, ArrowLeft } from 'lucide-react'
import Navbar from '../components/Navbar'
import Footer from '../components/Footer'

const NotFound = () => {
  const reduceMotion = useReducedMotion()

  return (
    <>
      <Navbar />
      <main className="relative flex min-h-[80vh] items-center justify-center px-5 pt-24 pb-16 sm:pt-28">
        <div className="mx-auto w-full max-w-3xl text-center">
```

```

<motion.div
  initial={{ opacity: 0, y: reduceMotion ? 0 : 24 }}
  animate={{ opacity: 1, y: 0 }}
  transition={{ duration: 0.65, ease: [0.22, 1, 0.36, 1] }}
>
  <span className="mb-6 inline-flex items-center gap-2
rounded-full border border-line-strong bg-parchment px-3.5 py-1.5 text-
xs font-semibold tracking-[0.14em] uppercase text-bronze-deep shadow-
soft">

    <Compass className="size-3.5" />
    Error 404
  </span>

  <div className="well mx-auto mb-6 inline-flex items-center
justify-center px-8 py-5 sm:px-12">
    <span className="font-display text-7xl font-bold tracking-
tight text-bronze sm:text-8xl">
      4
      <span className="mx-1 inline-block size-10 rounded-xl
border border-line-strong bg-cream align-baseline shadow-well sm:size-
14" />
      4
    </span>
  </div>

  <h1 className="font-display text-3xl font-bold text-ink
sm:text-4xl">
    This page drifted off the map
  </h1>
  <p className="mx-auto mt-4 max-w-xl text-sm leading-relaxed
text-muted sm:text-base">
    The page you&apos;re looking for doesn&apos;t exist or was
moved.

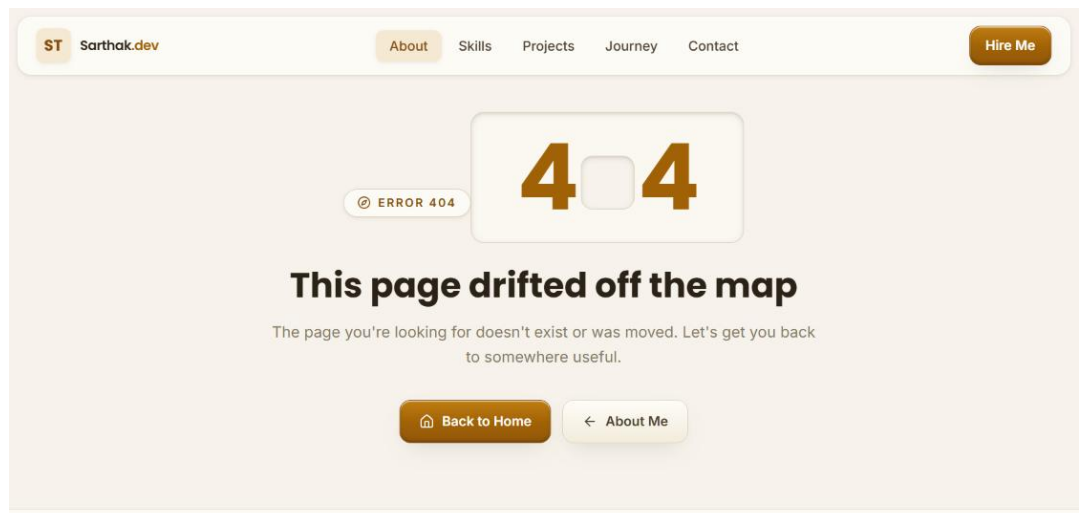
    Let&apos;s get you back to somewhere useful.
  </p>

  <div className="mt-8 flex flex-col items-center justify-
center gap-3 sm:flex-row">
    <Link
      to="/"
      className="btn-primary inline-flex cursor-pointer items-
center justify-center gap-2 rounded-xl px-5 py-3 text-sm font-semibold
transition-all duration-200 select-none hover:-translate-y-0.5"
    >
      <Home className="size-4" />
      Back to Home
    </Link>
    <Link
      to="/#about"

```

```
        className="btn-secondary inline-flex cursor-pointer
items-center justify-center gap-2 rounded-xl px-5 py-3 text-sm font-
semibold transition-all duration-200 select-none hover:-translate-y-0.5"
      >
        <ArrowLeft className="size-4" />
        About Me
      </Link>
    </div>
  </motion.div>
</div>
</main>
<Footer />
</>
)
}

export default NotFound
```

**Outcome**

Students will be able to implement client-side routing with React Router and manage dynamic UI state using `useState`, resulting in a multi-page application without full page reloads.